Question	Answer	Marks	AO	Guidance
16	$a\left(\frac{x_P + x_Q}{2}\right)^2 + b\left(\frac{x_P + x_Q}{2}\right) + c - a\left(\frac{x_P - x_Q}{2}\right)^2$ $= a\left\{\left(\frac{x_P + x_Q}{2}\right)^2 - \left(\frac{x_P - x_Q}{2}\right)^2\right\} + b\left(\frac{x_P + x_Q}{2}\right) + c$ $= a\left(\frac{x_P + x_Q}{2} + \frac{x_P - x_Q}{2}\right)\left(\frac{x_P + x_Q}{2} - \frac{x_P - x_Q}{2}\right) + b\left(\frac{x_P + x_Q}{2}\right) + c$ $= ax_Px_Q + b\left(\frac{x_P + x_Q}{2}\right) + c$	**M1** **A1**	**1.1** **2.1**	Dealing with the *a* terms convincingly by e.g. collecting terms with factor *a*, expanding the brackets and collecting etc. Squaring brackets or difference of two squares demonstrated to give a convincing completion to given result
		[2]		

Q16. To get the M1 they must work with the two terms with factor a. They may be moved next to each other, but they could also work with them in their original positions. The A1 is for manipulating the factor a terms correctly. This may be squaring of brackets or difference of two squares. But it must be convincing to get the required result. Give A0 for any error seen in the working.